Milestone 3 — Speaker Companion

Per-slide guidance, background, and Q&A backup

Deck spine (3 review comments). (1) SUTVA: the economic audit is the defense; the permutation test and HLT *confirm* it; break tests are informative diagnostics, not exclusion rules (`config.BREAKPOINT_EXCLUDE` is empty — pool follows the audit). **(2) Model behaviour:** behaviour = donor weights; sign the contamination bias (Hormuz conservative); robust to leave-one-out. **(3) Naive baselines:** freeze or over-extrapolate; the SCM is the data-driven counterfactual. **Headline:** Hormuz ∼59% (holds), Russia ∼22% (magnitude check).

1. How we defend donor selection (SUTVA) [60s]

Talking points

- **Primary defense is the economic audit** — the SCM-canonical approach (Abadie 2021 §3.1: donor exclusion rests on *substantive* grounds, not statistical pretests). Screen every candidate against two contamination channels: the oil-supply path (energy complex, petrocurrencies — categorically out) and the disruption itself (Hormuz routing, Russia/Ukraine trade — donor by donor).
- **Statistical cross-check — the two tests on the slides:**
  - **Permutation mean-shift** (Chow 1960 framing; Lehmann & Romano permutation reference) — the literal SUTVA question (did the donor's own series shift at T_0?), Brent-free, nonparametric. The validation doc designates this as the minimum load-bearing statistical confirmation, and it is the test that produces the lone flag (CNY).
  - **HLT trend break** — Brent-free and I(0)/I(1)-robust, so a break at T_0 *unambiguously* implicates the donor itself. The cleanest of the battery; flags nothing.
- **Bai–Perron and KS are diagnostics, not exclusion rules** — say this explicitly. Bai–Perron regresses on *treated* Brent, so a T_0 break is circular (could just be Brent moving); it measures *loading stability*, not SUTVA. KS over-rejects under macro-regime change. Both are reported but neither auto-excludes a donor.

The other at-T_0 test (run, not on the slides): Wilcoxon event-window

A third at-T_0 test sits in the battery but is kept off the slides:

- **Wilcoxon event-window** (Brown & Warner 1985, event-study tradition): compares the donor's returns in a *short window around* T_0 against a control window, rank-based. Where the permutation test looks for a *persistent mean shift* across the whole pre/post split, the event-window looks for a *localized return reaction* right at the event — they are complementary.
- **Why it is not shown:** on the retained pool it flags *nothing* — it does not even pick up CNY ($p \approx 0.44$ vs the permutation test's $p \approx 0.002$) — and it is lower-powered on the handful of event-window observations. So it adds no information to the result; the permutation test carries the lone flag on its own. Mention it only if asked "did you run more than one at-T_0 test?"
- In the combined BH-FDR flag the pipeline computes, the flag is "permutation *or* event-window"; since event-window fires on nothing here, the combined flag equals the permutation test alone.

Behind the slide

The hierarchy matters: audit is what a reviewer expects and what carries the argument; everything statistical is supplementary. We deliberately do *not* let a test exclude a donor automatically — the test flags a candidate, the researcher decides. This is the "report and interpret" convention of the SCM literature.

If asked

- *Why isn't Bai–Perron a SUTVA test?* It regresses each donor on Brent's returns. Brent is the treated unit, so at T_0 Brent itself jumps; a break in the donor–Brent relationship there cannot be separated from Brent's own move. It tells you whether the donor's *loading* is stable across the fitting window (a parallel-fit concern), not whether the donor was treated.
- *Why drop KS?* It rejects ∼21 of 32 donors on Russia — but that is the macro-regime change (COVID recovery → 2022 tightening), not per-donor treatment. Our own flag rule excludes it for that reason; it is a regime diagnostic, not a SUTVA test.
- *Is HLT definitive?* No — a non-rejection is failure-to-reject, not proof of no interference, and HLT only catches *trend* breaks. It is the cleanest *statistical* triangulation, but the audit remains primary.

2. SUTVA test: no retained donor is treated at T_0 [75s]

The table (verified, 19-donor retained pool)

Flagged at T_0 (of 19 retained)	Hormuz 2026	Russia 2022
HLT trend-break (Brent-free ⇒ SUTVA)	0 / 19	0 / 19
Permutation mean-shift *(on slides)*	0 / 19	1 / 19 (CNY)
Wilcoxon event-window *(run, not shown)*	0 / 19	0 / 19
Bai–Perron (loading stability; diagnostic)	0 / 19	1 / 19 (VIX)

Talking points

- **HLT flags nothing at either event** — the headline. Since it is Brent-free, this is the strongest clean read: no retained donor's own trend breaks at the event.
- **Two Russia flags, both retained on substantive grounds:**
  - **CNY** (permutation test) — the shift is the concurrent China zero-COVID reopening, not Russia-treatment. The test cannot separate the two; the audit judges it clean.
  - **VIX** (Bai–Perron) — a generic risk-off spike around the invasion, i.e. *intended* common-factor co-movement, which is what SCM exploits, not contamination.
- **Tests inform; the audit decides.** Neither flag triggers an exclusion — that is the policy, and it is why both donors stay in the 19-donor pool.

Behind the slide

VIX was *previously* excluded by an automatic "breakpoint rule." We removed that rule after two checks: (i) the Bai–Perron break does not reproduce when the test is re-run on the analysis window alone (it sits in the trimming dead zone; the contemporaneous early-2022 break is better attributed to HYG), and (ii) a VIX-in/out sensitivity moves the ensemble premium by only −0.2 pp (Russia) and +0.6 pp (Hormuz). Audit-vs-test agreement overall: Russia 18 / 32, Hormuz 31 / 32.

If asked

- *If a test flags VIX, why keep it?* The flag is a loading-stability signal, not SUTVA; it is sample-dependent (does not survive an analysis-window re-run); and it is immaterial to the estimate ($\leq$ 0.6 pp; see `scripts/vix_sensitivity.py`). Excluding it would also force dropping a clean, high-information donor from Hormuz purely to keep a shared pool.
- *Then why show Bai–Perron at all?* Honesty and completeness — it is a legitimate fit-stability diagnostic and it independently re-flags the audit's known cases (e.g. Wheat). We present it as

what it is, not as a SUTVA test.

- *Why does the permutation test flag CNY but not the audit?* The test sees "CNY moved at T_0"; it cannot attribute that to Russia versus the concurrent China reopening. The audit, which reasons about causal channels, places CNY's move on the China-policy channel — off the oil-supply / disruption path — so it stays.

- *Couldn't HLT just be under-powered on a short window?* On the full-history spec it is well-powered and clean. A window-restricted HLT does throw a few rejections, but at off-T_0 dates (COVID, 2025 metals moves) — volatility episodes, not focal-event treatment — so the full-history result is the right one to read.

3. Comment 2 — why the models behaved (donor weights) [~3 min, preface + 3 slides]

Spine (preface slide)

Behaviour = donor weights (which donors each model leans on) → *stress-test* (does any single donor drive the gap?) → *sign the contamination* (are those donors themselves moved by the treatment?). The preface states the method only; the conclusions come on the three content slides.

Talking points

- **Behaviour via donor weights** (heatmap). **Hormuz:** weight sits on safe-haven / non-oil donors (Gold, JPY, Sugar) that track common risk but *not* the oil-specific chokepoint shock ⇒ the synthetic undershoots Brent ⇒ the positive gap. **Russia:** the model spread is donor-driven — Convex/ASCM lean on Coffee & Platinum, XGBoost on SP500, and Bayesian Ridge degenerates ($w \approx 1.8$–2.4, outside the simplex) ⇒ dropped from the IQR.

- **Leave-one-out robustness.** Drop each >5%-weight donor, refit, report the ATT range. **Hormuz tight** (~6–10 pp on a ~59% effect) — no single donor flips it. **Russia wide** (14–27 pp); Bayesian Ridge crosses zero (LOO-fragile, consistent with its degenerate fit). Reinforces Russia-as-magnitude-check.

- **Signed contamination bias** (closing approach). Demand-destruction critique: high oil crowds out activity, so activity-proxy donors are pushed down ⇒ residual contamination. We decompose each linear model's gap (net bias $= -\sum_j w_j\,\delta_j$) and *sign* it. **Hormuz: negative** (-1.3 to -7.6%) ⇒ conservative, the ~59% is a *lower bound*. **Russia: positive** ($+12.7$ to $+21.4\%$; Bayesian Ridge -7.1) ⇒ overstate ⇒ Russia is a magnitude check.

Behind the slide

The Russia positive bias is the *same* demand/risk channel as Hormuz, just with the opposite net sign: the retained cyclical donors (Coffee, Platinum, SP500) were pushed *down* ($\delta < 0$), dragging the synthetic down and *inflating* the gap. (Do not say donors "drifted up" — that would imply a negative bias, the opposite of what we report.)

If asked

- *Which way does demand destruction bias the estimate?* Hormuz: toward zero (conservative) — so the headline is a floor. Russia: away from zero (the retained risk donors fell) — which is exactly why Russia is a magnitude check, not the headline.

- *Why is XGBoost absent from the bias table?* The additive $w_j\,\delta_j$ decomposition is only defined for the linear models; XGBoost is nonlinear, so its bias is not separable this way.

- *Why drop Bayesian Ridge from the IQR?* Degenerate convex weights ($w \approx 1.8$–2.4, far outside the simplex) and LOO-fragile (Russia range crosses zero). The ensemble IQR is the convex/ASCM/elastic/xgboost band; Bayesian Ridge is reported but down-weighted.

4. Comment 3 — naive counterfactual baselines [∼2 min, preface + 1 slide]

Talking points

- **The question:** ignore the donor pool and just extrapolate Brent's own past — what counterfactual do you draw, and how far off is the premium?
- **Two failure modes.** *Freeze / revert* baselines (flat & AR random walks, mean-reversion) understate — Hormuz ∼52%, Russia ∼9%. *Trend extrapolation* overshoots — linear pre-trend gives Hormuz **73.8%** (overstate); on Russia the extrapolated COVID-recovery rally overshoots actual Brent, *flipping the gap negative* (linear −4.2%, RW+drift −6.9%).
- **The SCM** alone tracks the data-driven, donor-implied path: **58.9%** Hormuz, **22.0%** Russia.
- **Plot note:** the slide shows the *entire* pre+post window — so the audience sees the full Russia COVID-recovery rally that the linear-trend / drift baselines extrapolate (and overshoot).

If asked

- *Why does the linear trend flip negative on Russia?* It extrapolates the pre-war rally; the implied counterfactual rises *above* observed Brent, so the gap goes negative. Naive extrapolation is spurious once the underlying factor turns.
- *Why is ARIMA(1,0,0) almost flat?* The best-AIC AR(1) has $\phi \approx 1$ — effectively a random walk — so its frozen forecast is near-flat (it *is* the ARIMA-based ITS counterfactual).

5. Summary — headline ATT [∼1 min]

Talking points

- **Hormuz ∼59%**: builds to its level by month 2–3 and holds (IQR $[54, 60]$) — the headline.
- **Russia ∼22%**: peaks ∼31% (1m) and settles ∼22% (full) — the magnitude check.
- Three legs hold it up: SUTVA (audit + tests), model behaviour (donor weights, signed bias, LOO), and the naive-baseline contrast.

If asked

- *Is the Russia effect statistically significant?* The in-space placebo (appendix) is *not* significant for Russia (mid-pack). Russia's case rests on the documented historical move ($95 → $130+), the audit-supported SUTVA defense, the cross-event weight transfer, and the conservative signed bias — not the single-event placebo. Hence "magnitude check," not headline.
- *Why is Hormuz the headline and Russia the check?* Russia's pre-period (2020-07 → 2022-02) spans COVID recovery + the reflation rally — high donor volatility weakens in-space inference; the Hormuz pre-period is calmer and the placebo is at the floor for every model.

6. Appendix (reserve) — in-space placebo [only if asked]

- **Treated-unit inference** (Abadie 2010): refit treating each donor as treated; rank Brent's post/pre RMSPE ratio against the $N = 20$ units (Brent + 19 donors). Floor $p = 1/20 = 0.05$.
- **Hormuz:** Brent ranks 1 under *all five* models $\Rightarrow p = 0.05$ (floor) — the strongest signal the pool allows. **Russia:** mid-pack (rank 8–12; XGBoost best at rank 4, $p = 0.20$) — not significant, consistent with the magnitude-check framing.
- **Why it is in the appendix:** it answers *significance*, not "why did the models behave" — a different question from Comment 2 — so it is reserve material for the significance Q.

7. Appendix (reserve) — walk-forward CV RMSE [only if asked]

- **What it is:** 5-fold expanding-window walk-forward CV (§5a). *val RMSE* = pooled out-of-sample error; *train RMSE* = cross-fold mean; *val/train* > 2 is a heuristic over-fit flag.

- **Result:** Russia **2/5** clean (Convex 1.25, Elastic-net 1.99 — ASCM 2.06, XGBoost 3.31, Bayesian Ridge 3.09 over-fit the volatile 2020–22 pre-period); Hormuz **4/5** clean (only Bayesian Ridge, 2.28).
- **Key point:** flagged models are *not* dropped — the ensemble takes the median, and the IQR band *is* the model-uncertainty. The ratio is a diagnostic, not a target.

8. Appendix (reserve) — per-model synthetic paths [only if asked]

- **What it shows:** observed Brent vs each of the five models' synthetic counterfactuals. Pre-T_0 the synthetics *track* Brent (the fit); post-T_0 they diverge — the gap is the premium.
- **Reading:** Convex SCM & ASCM are the headline pair; Bayesian Ridge's Russia path blows up (degenerate weights, w outside the simplex), which is exactly why it is down-weighted (cf. the LOO and RMSE appendices). Use this slide to make the divergence concrete if someone wants to "see the gap."